Project 2 Learner Guide

Santam Short-Term Insurance Underwriting and Claims Analytics

Why this project exists

Short-term insurance covers risks over defined policy periods, commonly renewed each year. Policies can change mid-term; premiums may be invoiced, paid, reversed or adjusted; and claims move through reporting, assessment, authorisation, repair and settlement. A single storm or flood can create many claims in one location over a short period.

These moving parts make a written-premium-only dashboard misleading. Your goal is to build a reproducible, auditable warehouse and show underwriting, claims, suspicious-pattern and geographic insights without turning a data-quality or fraud flag into an automatic decision.

Business unit in plain language

- Motor insurance covers insured vehicles and related risks, subject to policy wording, excesses and limits.
- Household insurance covers specified home contents or property risks. Area-level concentration matters when the same event affects many policies.
- Commercial insurance covers business property and other defined business risks. Individual losses can be much larger than an ordinary motor claim.
- Intermediaries include brokers, direct channels and bank-assurance. Production, commission and portfolio results should be compared on consistent definitions.
- Service providers can include assessors and repairers. Cost and speed need to be considered together with service quality and claim complexity.
- Claims have a lifecycle. A reported claim is not necessarily a settled claim, and paid value can differ from the authorised amount.

The earned-premium concept

The brief identifies earned premium as the technical heart of the project. Written premium describes a transaction or contract amount; earned premium allocates the amount to the days of risk actually provided during a reporting period. A policy covering 1 January through 31 December does not earn its entire annual premium on 1 January.

A simple pro-rata approach is:

earned premium for period = net written premium x covered days falling in period / total cover days

For monthly reporting, split the coverage interval at month boundaries. Define whether the cover end date is inclusive, how leap years are handled, and how mid-term endorsements or cancellations change the amount and coverage. A fully expired policy should reconcile to its net written premium under the agreed model, subject to rounding. Test a few policies by hand before running the full extract.

Core analytical questions

1. Underwriting profitability: Which product, channel and region combinations exceed a 100% combined ratio on an earned basis? Separate claim frequency from severity. Compare results with and without catastrophe-tagged claims.

2. Claims efficiency and leakage: Which providers combine slow settlement with high costs? How often is paid amount greater than authorised amount, and what assumptions support any annualised leakage estimate?

3. Suspicious-pattern indicators: Which claims merit human review because of timing, repetition, shared synthetic contact details, provider concentration or threshold-adjacent values? Explain false-positive risks.

4. Geographic concentration: Where is sum insured concentrated? Which dates show claim surges, and what loss might a hypothetical event produce under a clearly labelled damage-ratio assumption?

5. Data quality: Which defects dominate, how quickly are quarantines resolved, and what upstream control might prevent the most rework?

Each page should finish with a concise evidence-based headline and a proposed action. A risk flag is a prompt for qualified human review, not a fraud verdict, coverage decision or claim denial.

KPI definitions to start from

Earned premium - Baseline definition: Written premium x covered days in period / total cover days | Important decision: Define netting of credit notes, cancellations and endorsements; handle partial periods.

Loss ratio - Baseline definition: (Claims paid + change in outstanding reserves) / earned premium | Important decision: If reserves are absent, label it as a paid loss ratio and explain the limitation.

Expense ratio - Baseline definition: Commission + acquisition and administration expense / earned premium | Important decision: State which expense components are available in the extract.

Combined ratio - Baseline definition: Loss ratio + expense ratio | Important decision: Above 100% indicates the defined claims and expenses exceed earned premium, before any omitted items.

Claim frequency - Baseline definition: Claims reported / earned exposure in policy-years | Important decision: Define earned exposure and the unit of observation.

Average claim cost - Baseline definition: Total paid / number of settled claims | Important decision: Avoid dividing by all reported claims if many remain open.

Cycle time - Baseline definition: Settlement date - report date | Important decision: Report median and 90th percentile as well as an average; define open-claim treatment.

Leakage rate - Baseline definition: Positive amount of (paid - authorised) / total authorised | Important decision: A positive gap is a review signal, not proof of overpayment.

Fraud flag rate - Baseline definition: Claims above the documented score threshold / claims reported | Important decision: Report the threshold and explain that flags are not findings.

Concentration share - Baseline definition: Sum insured in area / total sum insured | Important decision: Define postal-area mapping, top-10 scope and any missing-geography treatment.

Claims and risk concepts

Accumulating-snapshot fact. A claim lifecycle table keeps one row per claim and fills milestone dates as the claim progresses: reported, assessed, authorised, repaired and settled. It is useful for cycle-time reporting and is updated as new events arrive. Preserve event history separately if you need a full audit trail.

Role-playing dates. A claim can have loss, report and settlement dates. Use a shared date dimension; the loss date is often the active relationship, with other date relationships activated in measures when needed. Test each measure against SQL because multiple date relationships can change filter behavior.

Type 2 history. When policy risk details change, keep a version with effective-from, effective-to and current-row flag rather than overwriting the old values. This lets you report exposure using the policy address, vehicle value or cover amount that applied on the event date. Ensure only one current version exists for each policy-risk entity.

Catastrophe accumulation. Many claims clustered in date and geography can signal a shared event, but correlation alone does not establish cause. Use the event calendar as a hypothesis, compare with nearby periods and areas, and state how event tagging was assigned.

Explainable review signals. A transparent score can add documented points for a claim within 30 days of inception, repeated claims, shared synthetic contact attributes, unusual provider/broker concentration or round amounts. Record each component separately, state why the threshold was selected, and evaluate false positives. Exclude protected or sensitive attributes and test whether location acts as an unfair proxy.

Data engineering concepts

Profile dates, claim states, money fields, premiums, risks and references before cleaning. The source files intentionally mix formats. A value that looks invalid may be a valid credit note or an event that is possible but unusual.

Use loss date <= report date <= settlement date as a starting chronology check, but do not rewrite dates without evidence. Check loss date against cover dates. Keep large, plausible catastrophe and commercial losses visible; do not cap genuine claims simply because they exceed an IQR fence. Apply caps only when a documented rule supports a data-entry error.

Resolve customer, vehicle and provider duplicates cautiously. Normalised synthetic identifiers or registrations can support deterministic matches. Fuzzy names and branches create candidates, not unquestionable truth. Preserve a crosswalk that can be reviewed and reversed.

For negative premium events, distinguish a matched cancellation or credit note from an unmatched suspicious entry. Reconcile amounts and references before classification. Track every correction in a POPIA-conscious audit log, and mask direct identifiers from reporting.

For every source extract, reconcile:

rows read = rows loaded + rows quarantined + intentionally dropped

Quarantine records need a reason code, source file, timestamp, status and reprocessing route. Repeated pipeline runs should not create duplicate premium, claim or audit rows.

Suggested anomaly catalogue

Use the BRD's starter codes and extend them when profiling identifies additional issues:

- B01 duplicate customer or risk: normalised ID, registration or VIN repeats; resolve with evidence and keep a crosswalk.
- B02 impossible vehicle year: after policy inception or before the chosen plausibility boundary; null or quarantine according to an explicit rule.
- B03 zero/extreme sum insured: investigate by line; cap a proven entry mistake or flag a plausible large exposure.
- B04 claim outside cover: loss before inception or after expiry; quarantine unless a documented correction is supported.
- B05 date-chain violation: report precedes loss or settlement precedes report; never silently reorder events.
- B06 paid above authorised / sum insured: flag for claims review with the relevant amount and policy version.
- B07 duplicate claim or transaction: matching policy, loss date, cause and amount; check legitimate split payments first.

- B08 orphan key: missing policy, customer, broker, vehicle, geography or provider parent.
- B09 provider variant: name/branch spelling differs from the reference; map with a logged alias.
- B10 unmatched credit note: negative premium has no matching original transaction reference.

A practical three-week sequence

Week 1 - Profile and define: agree the earned-premium method, build policy examples by hand, catalogue anomalies and create tested cleaning rules.

Week 2 - Model and reconcile: implement Type 2 policy-risk history, premium and claim facts, accumulating claim snapshots, quarantine, indexes and reconciliation views.

Week 3 - Analyse and communicate: build the six required reporting surfaces (executive summary plus five analytical tasks), verify measures against SQL, review flag fairness and prepare evidence-led actions.

Questions to defend in your final review

- How does a policy's earned premium change across its cover months?
- What happens to earned premium when the policy is endorsed or cancelled mid-term?
- Why is a high claim amount flagged rather than automatically capped?
- What false positives might your review indicators produce?
- How do Type 2 versions ensure an old claim uses the correct historical risk values?
- How would you distinguish an event cluster from a seasonal or reporting effect?
- What is excluded from the combined ratio because it is not present in the source data?

Short glossary

Accumulating snapshot: a fact row updated as a process reaches new milestones. Combined ratio: defined loss and expense costs divided by earned premium, commonly expressed as a percentage. Credit note: a transaction that reduces or reverses a prior premium charge. Earned exposure: insured time represented as a fraction of a policy-year. Endorsement: a mid-term policy change that may adjust cover and premium. Type 2 history: dimensional modelling that retains prior versions of changing attributes.

Training-data caution

All records are fictional and all identity-like/contact/banking values are synthetic placeholders. Do not use this sample to contact people, decide cover, settle claims or accuse anyone of fraud. A pattern indicator is for review only. The guide follows the supplied simulated requirements and does not replace legal, actuarial, claims or regulatory advice.